# Is Time Ticking?[†]

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**Abstract**

Continuous-time heterogeneous-agent models sidestep discrete-time Kuhn-Tucker conditions—but only when controls are instantaneous. Any decision with a temporal gap between commitment and payoff—time-to-build investment, contractual lock-in, durable purchases—restores these conditions: the agent must maintain liquidity throughout the gap, and this requirement reshapes policy functions, precautionary behavior, and the equilibrium wealth distribution. In a calibrated economy, the commitment channel reduces aggregate capital by up to 8%. Standard discrete time with period $\Delta t$ implicitly embeds such a gap; the continuous-time limit $\Delta t \to 0$ eliminates it. When the gap is structural, the limit does not refine the approximation—it removes the friction.

**Keywords**: Continuous time, discrete time, commitment frictions, time to build, Kuhn-Tucker conditions, heterogeneous agents.

**JEL Classification**: E21, E22, C61, D15.

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# 1 Introduction

A building under construction is a pure drain on the balance sheet: cash outflows are immediate, yet the half-built structure is neither habitable nor pledgeable. This temporal wedge between outlay and payoff—*irreversible commitment with delay*—pervades economic life, from time-to-build investment (Kydland and Prescott, 1982) to durable goods purchases, R&D, and infrastructure. A key advantage of continuous-time (CT) heterogeneous-agent models is that they sidestep the Kuhn-Tucker (KT) complementary slackness conditions of discrete-time (DT) formulations, because agents adjust on the margin at every instant (Achdou et al., 2022). This paper shows that the claim rests on an implicit assumption: all controls are instantaneous. Once agents face commitment frictions with temporal delay $\tau > 0$, KT conditions re-emerge within the CT formulation itself. A revealing way to see the issue is through the "+1" in discrete-time Euler equations, $u'(c_t) = \beta(1+r)\,\mathbb{E}[u'(c_{t+1})]$. The subscript $t+1$ serves two distinct roles: (a) a discretization artifact that vanishes as $\Delta t \to 0$; or (b) a genuine economic lag—commitment, gestation, or contractual lock-in—that should *not* vanish. The standard CT approach assumes all instances are of type (a). This paper shows that many economically important frictions are of type (b). Treating them as type (a) discards the Kuhn-Tucker multiplier that prices the commitment cost. Put differently: standard discrete time with period length $\Delta t$ implicitly embeds a time-to-build lag of $\Delta t$; the continuous-time limit $\Delta t \to 0$ eliminates this lag. If the lag is a discretization artifact, elimination is correct. If it reflects a structural friction—months to build a factory, quarters to process a mortgage—the CT model must restore it explicitly. Otherwise, the agent's need to maintain liquidity throughout the gap period, and the way that need reshapes decisions, will be absent from the model.

This paper makes two key contributions. *First*, the CT–DT modeling choice carries economic content beyond numerical convenience. Any decision with a temporal gap between commitment and payoff reintroduces KT conditions at the commitment point of the Hamilton-Jacobi-Bellman equation, because the agent must maintain liquidity throughout the gap. The delay parameter $\tau$ indexes a continuous family interpolating between instantaneous adjustment ($\tau = 0$) and commitment with gestation ($\tau > 0$). Standard DT implicitly embeds a friction of duration $\Delta t$ that the CT limit eliminates. A three-date example delivers this result in closed form; the gen-

eral CT formulation confirms it. *Second*, incorporating the commitment channel into the CT framework has first-order quantitative effects. In a calibrated heterogeneous-agent economy, agents who anticipate the liquidity drain of ongoing projects invest less. This precautionary under-investment channel lowers the stationary capital stock by up to 8% for empirically relevant gestation lags, raises the equilibrium interest rate by 79 basis points, and reshapes the wealth distribution.

**Related literature** This paper connects to four strands of literature. The first is the CT heterogeneous-agent framework of Achdou et al. (2022), extended by Kaplan et al. (2018) and Moll (2014) to monetary policy and capital misallocation, respectively. These contributions treat all controls as instantaneous; the present paper complements this framework by incorporating controls with temporal delay, showing that KT conditions re-emerge at the commitment point.

The second is the time-to-build and irreversible investment literature. Kydland and Prescott (1982) introduce time-to-build as a propagation mechanism, Bar-Ilan and Strange (1996) study investment lags in partial equilibrium, Abel and Eberly (1996) characterize optimal investment under costly reversibility, and Dixit and Pindyck (1994) provide a comprehensive treatment of investment under uncertainty. More recently, Khan and Thomas (2008) and Winberry (2021) study lumpy investment with nonconvex adjustment costs in heterogeneous-firm models, and Baley and Blanco (2026) show that irreversibility—a *price* wedge between buying and selling capital—doubles the persistence of aggregate capital fluctuations. The present paper studies a complementary friction: a *temporal* wedge between commitment and payoff, and traces its implications for the optimality conditions in the CT heterogeneous-agent setting.

The third is the DT heterogeneous-agent tradition of Bewley (1986), Huggett (1993), and Aiyagari (1994), where Carroll (1997) and Krusell and Smith (1998) emphasize precautionary saving under income risk. The commitment friction studied here provides a distinct source of precautionary behavior—one operating through the liquidity cost of locked-up investment rather than income uncertainty alone. The paper also connects to the $(S, s)$ adjustment literature (Stokey, 2009; Grossman and Laroque, 1990; Alvarez and Lippi, 2014), where agents face fixed costs of adjustment and act infrequently. The commitment friction differs from a fixed cost: it is the temporal delay, not the adjustment cost, that generates the KT multiplier. Yet the

resulting inaction regions share a common mathematical structure.

Finally, I draw on the mathematical literature on delay differential equations and HJB equations in infinite-dimensional spaces (Fabbri et al., 2017; Bambi, 2008) and on convergence of approximation schemes for nonlinear equations (Barles and Souganidis, 1991). The delay $\tau$ transforms the HJB equation into a problem with history dependence; the analytical results characterize how KT conditions manifest in this richer setting.

# 2 A Three-Date Example

Before turning to the general continuous-time model, I use the simplest possible environment to show what the CT limit discards. Three dates, one agent, no uncertainty—the only difference between the two models below is *when* the investment return arrives. This is enough to generate a closed-form divergence that previews the full result.

## 2.1 Environment

There are three dates, $t \in \{0, 1, 2\}$. A single agent has log utility $u(c) = \ln c$ and discount factor $\beta \in (0, 1)$. Lifetime utility is

$$U = \ln c_0 + \beta \ln c_1 + \beta^2 \ln c_2. \tag{1}$$

**Endowments.** The agent enters date 0 with wealth $W > 0$. At date 1 she receives income $y > 0$. There is no endowment at date 2.

**Technology.** The agent chooses investment $i \geq 0$ at date 0. Production is $f(i) = Ai^\alpha$ with $A > 0$ and $\alpha \in (0, 1)$.

**Savings.** A risk-free storage technology carries resources forward at gross return 1. Liquid savings $s_t \geq 0$ at each date.

I study two models that differ *only* in the timing of the investment return.

- **Model F** (Flexible): Investment $i$ is paid at $t = 0$ and returns $f(i)$ at $t = 1$.

- **Model C** (Commitment): Investment $i$ is paid at $t = 0$ and returns $f(i)$ at $t = 2$.

Model C introduces a one-period *build lag*: resources committed at date 0 are locked up until date 2. The agent cannot access the return at the intermediate date. In the language of the Introduction, Model F corresponds to the CT formulation: investment returns are available immediately for consumption smoothing. Model C corresponds to DT with a one-period gestation lag: the agent must bridge a gap period before the return arrives.

## 2.2 Budget Constraints

**Model F.**

$$t = 0: \quad c_0 + i + s_0 = W, \tag{2}$$

$$t = 1: \quad c_1 + s_1 = s_0 + y + f(i), \tag{3}$$

$$t = 2: \quad c_2 = s_1. \tag{4}$$

**Model C.**

$$t = 0: \quad c_0 + i + s_0 = W, \tag{5}$$

$$t = 1: \quad c_1 + s_1 = s_0 + y, \tag{6}$$

$$t = 2: \quad c_2 = s_1 + f(i). \tag{7}$$

The only difference is the location of $f(i)$: it appears at $t = 1$ in Model F and at $t = 2$ in Model C. In both cases, combining the period budgets yields an identical *lifetime* budget constraint:

$$c_0 + c_1 + c_2 = W + y + f(i) - i. \tag{8}$$

Total lifetime resources are the same for any given $i$. The models disagree only on *when* those resources are available. If the savings constraints $s_0 \geq 0$ and $s_1 \geq 0$ never bind, the two models are equivalent. The question is whether, and when, those constraints do bind.

## 2.3 Model F: No Commitment Friction

In Model F, substitute the budget constraints and ignore the non-negativity constraints (verified shortly). The problem reduces to choosing $i$ and splitting lifetime resources $W + y + f(i) - i$ across three dates. With log utility and gross return 1, the Euler equations $u'(c_t) = \beta u'(c_{t+1})$ imply

$$c_0 : c_1 : c_2 = 1 : \beta : \beta^2. \tag{9}$$

The consumption allocation is therefore

$$c_0^F = \frac{W + y + f(i) - i}{1 + \beta + \beta^2}, \quad c_1^F = \frac{\beta\big[W + y + f(i) - i\big]}{1 + \beta + \beta^2}, \quad c_2^F = \frac{\beta^2\big[W + y + f(i) - i\big]}{1 + \beta + \beta^2}. \tag{10}$$

The optimal investment maximizes $W + y + f(i) - i$, which requires

$$f'(i^F) = \alpha A (i^F)^{\alpha - 1} = 1 \quad \Longrightarrow \quad i^F = (\alpha A)^{\frac{1}{1-\alpha}}. \tag{11}$$

This is the *technology-determined* investment level: it equates the marginal product of investment to the gross return on storage. Crucially, $i^F$ depends on $A$ and $\alpha$ but *not on* $W$. Wealth affects only the level of consumption, not the investment decision.

To verify that no constraint binds, note that $s_0 = W - c_0^F - i^F$ and $s_1 = s_0 + y + f(i^F) - c_1^F$. For $W$ sufficiently large, both are positive and the complete-markets allocation obtains.[1]

## 2.4 Model C: Commitment Friction

In Model C, the investment return $f(i)$ arrives at $t = 2$, one period after the agent might need it. The Lagrangian is

$$\begin{aligned} \mathcal{L} = {} & \ln c_0 + \beta \ln c_1 + \beta^2 \ln c_2 \\ & + \lambda_0\big(W - c_0 - i - s_0\big) + \lambda_1\big(s_0 + y - c_1 - s_1\big) + \lambda_2\big(s_1 + f(i) - c_2\big) \\ & + \mu_0\, s_0 + \mu_1\, s_1, \end{aligned} \tag{12}$$

[1] In Model F, if $W$ is too small to fund both $i^F$ and the desired $c_0^F$, then $s_0 \geq 0$ binds and the agent underinvests. But because $f(i)$ arrives at $t = 1$ — immediately after the investment is paid — the date-1 and date-2 constraints remain slack. The constrained region in Model F arises only from a standard funding constraint, not from a timing mismatch. This distinction is central to the comparison with Model C.

where $\lambda_t$ are the multipliers on the period budget constraints and $\mu_t \geq 0$ are the multipliers on $s_t \geq 0$.

The first-order conditions for consumption yield

$$\frac{1}{c_0} = \lambda_0, \quad \frac{\beta}{c_1} = \lambda_1, \quad \frac{\beta^2}{c_2} = \lambda_2. \tag{13}$$

For savings,

$$\lambda_0 = \lambda_1 + \mu_0, \quad \lambda_1 = \lambda_2 + \mu_1. \tag{14}$$

For investment,

$$\lambda_0 = \lambda_2 f'(i). \tag{15}$$

**Which constraint binds?** The economic logic is immediate. In Model C, the agent pays $i$ at date 0 but receives $f(i)$ only at date 2. Date 1 is the *gap period*: resources flow out (savings from date 0) and in (income $y$), but the investment return is absent. If the agent could borrow at date 1 against the pipeline return $f(i)$ arriving at date 2, she would. But borrowing requires $s_1 < 0$, which the savings constraint prohibits. It is the *date-1 constraint* — the gap-period constraint — that threatens to bind, not the date-0 constraint.

This is the central insight: commitment friction is not about whether the agent can afford to invest (a date-0 issue), but about whether she can sustain consumption *while waiting for the return* (a date-1 issue).

**Three regimes.** The solution has three regimes, depending on initial wealth $W$.

**Regime 0: Unconstrained** ($W$ large). Neither $s_0 \geq 0$ nor $s_1 \geq 0$ binds. Then $\mu_0 = \mu_1 = 0$, so $\lambda_0 = \lambda_1 = \lambda_2$ and the Euler equations hold with equality throughout. The solution coincides with Model F: the investment rule is $f'(i) = 1$ and consumption is spread in the ratio $1 : \beta : \beta^2$. Models F and C agree whenever the agent is wealthy enough that borrowing constraints are slack.

In the unconstrained solution of Model C, date-1 savings are $s_1 = s_0 + y - c_1$, where $f(i)$ is absent (it arrives at $t = 2$). Substituting the unconstrained consumption and investment rules, $s_1 \geq 0$ holds if and only if

$$W + y - i^F \;\geq\; \frac{(1+\beta)\big[W + y + f(i^F) - i^F\big]}{1 + \beta + \beta^2}. \tag{16}$$

Rearranging, there exists a finite threshold $\bar{W}$ such that the models agree if and only if $W \geq \bar{W}$:

$$\bar{W} = i^F + \frac{(1+\beta)f(i^F)}{\beta^2} - y. \tag{17}$$

Below $\bar{W}$, the gap-period constraint binds and the models diverge.

**Regime 1: Gap-period constrained** ($W$ intermediate). The constraint $s_1 \geq 0$ binds ($\mu_1 > 0$) but $s_0 \geq 0$ remains slack ($\mu_0 = 0$). This is the *key case*. Because $\mu_0 = 0$, the Euler equation between dates 0 and 1 holds:

$$\frac{1}{c_0} = \frac{\beta}{c_1} \quad \Longrightarrow \quad c_1 = \beta\, c_0. \tag{18}$$

Because $s_1 = 0$ binds, the date-2 budget constraint gives

$$c_2 = f(i). \tag{19}$$

The agent's date-2 consumption is *exactly* the investment return — nothing more, nothing less. She cannot smooth between dates 1 and 2: all date-1 resources are consumed at date 1, and the only date-2 resource is the pipeline return.

The binding constraint $s_1 = 0$ together with the date-1 budget gives $c_1 = s_0 + y$, and $s_0 = W - c_0 - i$, so

$$c_1 = W - c_0 - i + y. \tag{20}$$

Combining with $c_1 = \beta c_0$ from (18):

$$c_0 = \frac{W + y - i}{1+\beta}. \tag{21}$$

Now substitute into the investment first-order condition. From (15) with $\mu_0 = 0$:

$$\lambda_0 = \lambda_2 f'(i) \quad \Longrightarrow \quad \frac{1}{c_0} = \frac{\beta^2 f'(i)}{c_2} = \frac{\beta^2 \alpha A i^{\alpha-1}}{A i^{\alpha}} = \frac{\beta^2 \alpha}{i}. \tag{22}$$

Using $c_0 = (W + y - i)/(1+\beta)$:

$$\frac{1+\beta}{W + y - i} = \frac{\beta^2 \alpha}{i}. \tag{23}$$

Solving for $i$:

$$i^C = \frac{\beta^2\alpha}{1+\beta+\beta^2\alpha}(W+y). \tag{24}$$

This is the central result of the example. In Model F, the investment rule $i^F = (\alpha A)^{1/(1-\alpha)}$ is pinned down entirely by technology parameters. In Model C, the investment rule (24) is *linear in wealth*. The commitment friction has transformed a technology condition into a liquidity condition.

**Regime 2: Doubly constrained** ($W$ very low). Both $s_0 = 0$ and $s_1 = 0$ bind. The agent consumes all available resources at each date: $c_0 = W - i$, $c_1 = y$, $c_2 = f(i)$. The Euler equation between dates 0 and 1 no longer holds. Investment is determined by equating the marginal utility cost of investing (foregone date-0 consumption) to the discounted marginal utility of the investment return at date 2:

$$\frac{1}{W-i} = \frac{\beta^2 f'(i)}{f(i)} = \frac{\beta^2\alpha}{i}. \tag{25}$$

Solving: $i = \beta^2\alpha\, W/(1+\beta^2\alpha)$. Investment is again proportional to wealth but now independent of income $y$, since all income is consumed at date 1 and the two sides of the gap are completely disconnected.

## 2.5 The Divergence

**Proposition 2.1** (Qualitative Divergence)**.** *In Model F, the optimal investment*

$$i^F = (\alpha A)^{\frac{1}{1-\alpha}}$$

*is determined by technology alone and is independent of initial wealth $W$. In Model C with the gap-period constraint binding (Regime 1), the optimal investment*

$$i^C = \frac{\beta^2\alpha}{1+\beta+\beta^2\alpha}(W+y)$$

*is a linear function of wealth. The commitment friction transforms the investment rule from a technology condition to a wealth-dependent liquidity rule.*

*Proof.* Equations (11) and (24), derived above. ■

The proposition identifies the paper's core mechanism. In Model F, the investment

decision separates from the consumption-savings decision: the agent first chooses $i$ to maximize the surplus $f(i) - i$, then splits that surplus across dates. This separation breaks down in Model C because the commitment lag introduces a period during which the agent must finance consumption without access to the investment return. The liquidity cost of bridging this gap falls on investment, making the investment decision dependent on the agent's wealth.

## 2.6 The Shadow Value of the Gap Period

The Kuhn-Tucker multiplier $\mu_1$ on the gap-period constraint $s_1 \geq 0$ has a direct economic interpretation. In Regime 1, $\mu_1 = \lambda_1 - \lambda_2$, which from the first-order conditions (13) gives

$$\mu_1 = \frac{\beta}{c_1^C} - \frac{\beta^2}{c_2^C} = \frac{\beta}{c_1^C} - \frac{\beta^2}{f(i^C)} > 0. \tag{26}$$

The multiplier measures the *shadow value of being able to borrow at date 1 against the pipeline return $f(i)$ arriving at date 2.* It is the wedge between the marginal value of a unit of resources at date 1 (the gap period) and its discounted marginal value at date 2. In Model F, $\mu_1 = 0$ and resources flow freely across the investment horizon. In Model C, $\mu_1 > 0$ and the agent would pay a premium for date-1 liquidity.

Two observations follow immediately. First, $\mu_1$ is decreasing in $W$: wealthier agents face a smaller liquidity wedge because they can self-insure through larger initial savings $s_0$. Second, $\mu_1 \to 0$ as $W \to \bar{W}$ from below: the commitment friction vanishes continuously as wealth rises to the unconstrained threshold.

## 2.7 Comparative Statics

The investment wedge $i^F - i^C$ in Regime 1 summarizes the quantitative bite of the commitment friction.

**Wealth.** Since $i^C$ is linear in $W$ while $i^F$ is constant, the wedge $i^F - i^C$ is *decreasing* in $W$:

$$\frac{\partial(i^F - i^C)}{\partial W} = -\frac{\beta^2\alpha}{1 + \beta + \beta^2\alpha} < 0. \tag{27}$$

Richer agents invest closer to the first-best. This is intuitive: wealth provides self-insurance against the gap-period illiquidity, and more of it reduces the distortion.

**Income.** The wedge is also *decreasing* in $y$:

$$\frac{\partial(i^F - i^C)}{\partial y} = -\frac{\beta^2\alpha}{1+\beta+\beta^2\alpha} < 0. \tag{28}$$

Income $y$ arrives precisely during the gap period and directly relaxes the binding constraint. Higher date-1 income substitutes for borrowing against the pipeline return.

**Convergence.** As $W + y$ grows, $i^C \to i^F$ from below. The two models agree in the limit of abundant liquidity. The commitment friction matters when the agent is liquidity-constrained relative to the scale of her investment — precisely the empirically relevant case for entrepreneurs, firms with large capital expenditures, and households making durable purchases.

## 2.8 Quantitative Analysis

To make the magnitudes concrete, set $\beta = 0.95$, $A = 2$, $\alpha = 0.5$, and $y = 1$. Under these parameters, the first-best investment is $i^F = (\alpha A)^{1/(1-\alpha)} = 1.00$, and the Regime 1 coefficient on wealth in the commitment model is $\beta^2\alpha/(1+\beta+\beta^2\alpha) \approx 0.188$. The unconstrained threshold for Model C is $\bar{W} \approx 4.32$ from (17).

Table 1 reports the optimal allocations for both models as $W$ varies from 1.0 to 5.0. For $W \geq 3$, Model F achieves the first-best investment $i^F = 1.00$, while Model C continues to underinvest until $W$ exceeds $\bar{W}$.

The most revealing rows are $W = 3$ and $W = 4$, where Model F achieves the first-best but Model C does not. At $W = 3$, the commitment model invests only 75% of the efficient level. The agent would like to invest more but cannot bridge the gap period without running out of liquid resources at date 1. At $W = 4$, the gap has nearly closed — the agent is wealthy enough to self-insure through the build lag. By $W = 5$, the models coincide: wealth is large enough that the savings constraints are slack and the commitment lag is irrelevant.

Even at $W = 1$ and $W = 2$, where both models are constrained, the *nature* of the constraint differs. In Model F, the binding constraint is $s_0 \geq 0$ — a standard funding constraint at the date of investment. In Model C, the binding constraint is $s_1 \geq 0$ — a liquidity constraint at the gap period, one step removed from the investment date. The commitment friction shifts the locus of the constraint from the point of expenditure to the period of waiting.

Table 1: Three-date example: Model F vs. Model C

| | Investment | | $c_0$ | | $c_2$ | | Multiplier |
|---|---|---|---|---|---|---|---|
| $W$ | $i^F$ | $i^C$ | $c_0^F$ | $c_0^C$ | $c_2^F$ | $c_2^C$ | $\mu_1$ |
| 1.0 | 0.33 | 0.31 | 0.67 | 0.69 | 1.05 | 1.12 | 0.14 |
| 2.0 | 0.74 | 0.56 | 1.26 | 1.25 | 1.32 | 1.50 | 0.20 |
| 3.0 | 1.00 | 0.75 | 1.75 | 1.67 | 1.58 | 1.73 | 0.08 |
| 4.0 | 1.00 | 0.94 | 2.10 | 2.08 | 1.90 | 1.94 | 0.02 |
| 5.0 | 1.00 | 1.00 | 2.45 | 2.45 | 2.22 | 2.22 | 0.00 |

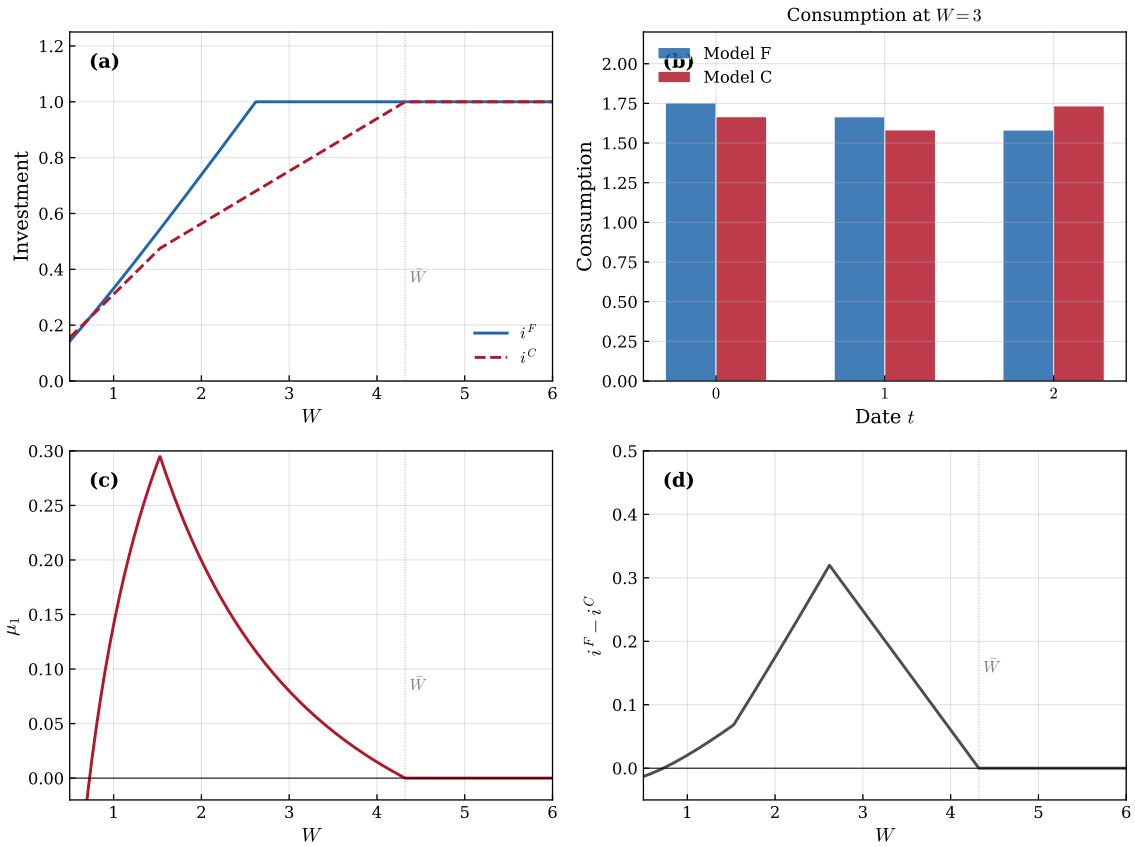
*Notes*: Parameters: $\beta = 0.95$, $A = 2$, $\alpha = 0.5$, $y = 1$. Model F investment is $i^F = (\alpha A)^{1/(1-\alpha)} = 1.00$ when $W$ is large enough that the date-0 savings constraint is slack; for low $W$, Model F is also constrained but by a standard funding constraint at $t = 0$. Model C investment $i^C$ is given by (24) in Regime 1 (gap-period constrained). The multiplier $\mu_1$ is the shadow value of relaxing the date-1 savings constraint in Model C.

Figure 1 displays four panels. Panel (a) plots $i^F$ and $i^C$ against $W$, showing the constant first-best investment (for large $W$) and the linearly increasing commitment investment that converges from below. Panel (b) shows the consumption paths $\{c_0, c_1, c_2\}$ for each model at $W = 3$, illustrating how the commitment friction compresses date-1 consumption and inflates date-2 consumption relative to the first-best. Panel (c) plots the Kuhn-Tucker multiplier $\mu_1$ as a function of $W$, showing its decline to zero at the unconstrained threshold $\bar{W}$. Panel (d) plots the investment wedge $i^F - i^C$ against $W$.

## 2.9 Lesson

The example makes visible what the continuous-time limit discards. Moving from Model C to Model F—the three-date analog of taking $\Delta t \to 0$—eliminates the gap period. With it goes the constraint that forces the agent to maintain liquidity while waiting for the return. The Kuhn-Tucker multiplier $\mu_1$ vanishes, the investment rule reverts to technology alone, and the role of wealth disappears. In three dates this is a modeling choice between two simple problems. In continuous time, it is the default: the standard CT formulation assumes $\tau = 0$, so the gap period is absent from the model. The remainder of the paper shows how to restore it—and that the quantitative consequences of doing so are substantial.

Figure 1: Three-date example: Investment, consumption, and the commitment wedge




# 3 Baseline Benchmark: Instantaneous Controls

This section establishes the benchmark: for the canonical consumption-savings problem with instantaneous controls, the discrete-time KT structure vanishes in the continuous-time limit (Achdou et al., 2022). The result serves as a null hypothesis against which the commitment friction results of Section 4 can be measured.

## 3.1 Discrete Time

Consider the standard Bewley (1986)–Huggett (1993)–Aiyagari (1994) consumption-savings problem. Time is discrete with period length $\Delta t$. An agent with wealth $a_t$ chooses consumption $c_t$ to solve

$$V(a) = \max_{c \geq 0} \left\{ u(c)\, \Delta t + \beta\, \mathbb{E}[V(a')] \right\}, \tag{29}$$

subject to the budget constraint and borrowing limit,

$$a' = (1 + r\,\Delta t)\,a + w\,\Delta t - c\,\Delta t, \qquad a' \geq \underline{a}. \tag{30}$$

Here $r$ is the interest rate, $w$ is labor income (possibly stochastic), and $\beta = e^{-\rho\,\Delta t}$ is the discount factor corresponding to discount rate $\rho$. The utility function $u$ is strictly concave, strictly increasing, and satisfies the Inada conditions.

The first-order and complementary slackness conditions are

$$u'(c_t) = \beta\,(1 + r\,\Delta t)\,\mathbb{E}[u'(c_{t+\Delta t})] + \mu_t, \tag{31}$$

$$\mu_t \geq 0, \qquad \mu_t(a_{t+\Delta t} - \underline{a}) = 0, \tag{32}$$

where $\mu_t$ is the Kuhn-Tucker multiplier on the borrowing constraint. When the constraint binds ($a_{t+\Delta t} = \underline{a}$), the multiplier $\mu_t > 0$ drives a wedge between current marginal utility and the discounted expected future marginal utility. This wedge is the source of the well-known precautionary savings channel. Because $u''' > 0$ (prudence) and there is a positive probability of hitting the constraint in the future, the agent saves more than the certainty-equivalent level even when the constraint does not currently bind.

## 3.2 Continuous Time

The continuous-time formulation follows Achdou et al. (2022). The Hamilton-Jacobi-Bellman equation is

$$\rho\,V(a) = \max_{c \geq 0} \left\{ u(c) + V'(a)(r\,a + w - c) \right\}, \tag{33}$$

with the state constrained to $a \geq \underline{a}$. The first-order condition is

$$u'(c) = V'(a). \tag{34}$$

There is no Kuhn-Tucker multiplier in equation (34). Instead, the borrowing constraint is absorbed into a *boundary condition* on the value function at $a = \underline{a}$: the drift

of wealth must be non-negative at the boundary,

$$r\,\underline{a} + w - c(a)\big|_{a=\underline{a}} \geq 0, \tag{35}$$

which pins down the consumption policy at the constraint. The agent can never overshoot below $\underline{a}$ because the control adjusts instantaneously to prevent this.

The absence of a multiplier in the continuous-time FOC is the key simplification emphasized by Achdou et al. (2022). One solves a smooth HJB equation on $[\underline{a}, \infty)$ with a reflecting boundary, rather than handling complementary slackness at each state.

## 3.3 Reconciliation: the Limit $\Delta t \to 0$

The discrete-time formulation (29)–(32) converges to the continuous-time formulation (33)–(35) as $\Delta t \to 0$. The formal convergence is standard and follows from the general results in Kushner and Dupuis (2001) and Barles and Souganidis (1991). The economic intuition is straightforward:

(i) **Step size vanishes.** As $\Delta t \to 0$, the change in wealth over one period, $a_{t+\Delta t} - a_t = (r\,a_t + w - c_t)\,\Delta t$, shrinks to zero. The probability that a single period's transition overshoots the borrowing constraint from any interior point $a_t > \underline{a}$ vanishes.

(ii) **Multiplier vanishes.** Since the constraint is reached only in the limit (via the boundary condition on the drift), the Kuhn-Tucker multiplier $\mu_t$ in (31) converges to zero. In discrete time, $\mu_t$ compensates for the possibility that a finite step might push $a_{t+\Delta t}$ to $\underline{a}$; as the step size shrinks, this compensation becomes unnecessary.

(iii) **Precautionary motive from constraint overshooting vanishes.** In the discrete-time Euler equation, the multiplier $\mu_t$ generates a precautionary savings wedge even at interior points, because the agent internalizes the probability of hitting the constraint within one period. This channel disappears in continuous time.

(iv) **Remaining precautionary motive survives.** The precautionary savings motive arising from prudence ($u''' > 0$) and income risk — operating through

Jensen's inequality applied to $\mathbb{E}[u'(c_{t+\Delta t})]$ — is present in both formulations and is unaffected by the limit.

In sum, the two models agree because the Kuhn-Tucker multiplier is a discrete-time artefact that vanishes in the limit. The continuous-time boundary condition is sufficient to capture the constraint. The claim in Achdou et al. (2022) is correct in this setting.

### 3.4 Preview: Where the Equivalence Breaks

The equivalence established above relies on a critical feature: consumption is an *instantaneous control*. The agent can always reduce consumption just enough to prevent wealth from crossing the borrowing constraint; there is no delay between the decision and its effect on the state. This instantaneous adjustability is what allows the boundary condition to replace Kuhn-Tucker without loss.

This equivalence breaks when the agent faces a *commitment friction* — a control that, once chosen, locks in resources for a duration $\tau > 0$ before the outcome is realized. The agent cannot fine-tune the control to avoid the constraint during the gap, and the Kuhn-Tucker multiplier re-emerges. Any continuous-time model in which a key decision has positive gestation—investment, hiring, durable purchases—therefore inherits the constraint structure that the boundary-condition approach was designed to eliminate.

## 4 Continuous Time with Commitment Frictions

I now introduce a continuous-time model in which investment is subject to a commitment friction with temporal delay $\tau > 0$. Consumption remains a continuous flow control, but investment is chosen at discrete commitment points. The HJB equation inherits Kuhn-Tucker conditions that cannot be absorbed into boundary conditions.

### 4.1 Setup and the Four-Channel Exposure

Consider an agent with three individual state variables: liquid assets $a_t$, installed capital $k_t$, and idiosyncratic productivity $z_t$. The aggregate state consists of the

aggregate capital stock $K_t$ and an aggregate productivity process $Z_t$. The laws of motion are

$$\mathrm{d}a_t = \big[r_t\, k_t + w_t\, z_t - c_t - i_t\big]\mathrm{d}t + \sigma_a\, \mathrm{d}W_t^a, \tag{36}$$

$$\mathrm{d}k_t = \big[i(t-\tau) - \delta\, k_t\big]\mathrm{d}t, \tag{37}$$

$$\mathrm{d}z_t = m_z(z_t)\,\mathrm{d}t + \sigma_z(z_t)\,\mathrm{d}W_t^z, \tag{38}$$

$$\mathrm{d}Z_t = m_Z\,\mathrm{d}t + \sigma_Z\,\mathrm{d}W_t^Z. \tag{39}$$

The rental rate $r_t = r(K_t, Z_t)$ is the net return on capital (gross marginal product minus depreciation). The wage $w_t = w(K_t, Z_t)$ is the marginal product of labor. Both are determined by the aggregate state. Note that depreciation in (37) governs the evolution of the agent's *installed* capital stock, while $r_t$ in (36) is the net rental rate on that capital. The borrowing constraint requires $a_t \geq \underline{a}$ at all times, and investment is irreversible: $i_t \geq 0$.

**Assumption 4.1** (Non-pledgeability of pipeline investment)**.** Investment in the pipeline—committed at date $t$ but not yet installed at $t+\tau$—cannot be used as collateral and does not relax the borrowing constraint $a_t \geq \underline{a}$ during the gestation interval $[t, t+\tau]$.

Assumption 4.1 captures the idea that half-built capital is illiquid: a partially completed factory cannot be sold or pledged at full value. This is the source of the commitment cost. If pipeline investment were fully pledgeable, the borrowing constraint would never tighten during gestation and the KT multiplier would vanish. The baseline assumption of zero pledgeability is conservative; Section 7 discusses robustness to partial pledgeability.

The critical feature is equation (37): investment committed at date $t$ augments installed capital only at date $t+\tau$, where $\tau > 0$ is the time-to-build delay.[2]

At the moment the agent commits investment $i_t$, the agent's state is $(a_t, k_t, z_t)$ and the aggregate state is $(K_t, Z_t)$. The prices at which the commitment is made are $(r_t, w_t)$. But the payoff from the investment—the increase in installed capital and its subsequent income stream—is not realized until $t+\tau$. During the interval $[t, t+\tau]$, the agent is exposed to changes along four distinct channels.

[2]The specification follows Kydland and Prescott (1982) and the continuous-time time-to-build literature (Bambi, 2008; Fabbri et al., 2017). See Online Appendix A for the full delay differential equation formulation.

**Channel 1: Exogenous idiosyncratic exposure.** The idiosyncratic productivity process $z_s$ evolves stochastically over $[t, t+\tau]$ according to (38). An adverse productivity shock during gestation tightens the agent's liquidity position, but the committed investment cannot be revised. Under instantaneous investment ($\tau = 0$) the agent adjusts at the next instant; with $\tau > 0$, the response is delayed by the full gestation period.

**Channel 2: Endogenous idiosyncratic exposure.** Prior investment commitments mature and current capital depreciates during $[t, t+\tau]$. The capital income flow $r_s k_s$ changes in ways shaped by both past decisions and the evolution of prices, creating a feedback loop between the agent's prior commitments and her state at $t+\tau$.

**Channel 3: Exogenous aggregate exposure.** Aggregate productivity $Z_s$ evolves over $[t, t+\tau]$. The agent committed $i_t$ at date-$t$ prices $(r_t, w_t)$, but these are not the prices that prevail during gestation or at the date of capital arrival. The commitment is a bet that aggregate conditions will not deteriorate before the investment matures.

**Channel 4: Endogenous aggregate exposure.** The aggregate capital stock $K_s$ evolves as commitments of all agents mature, altering prices $r_s = r(K_s, Z_s)$ and $w_s = w(K_s, Z_s)$. In general equilibrium, the interaction between individual commitment decisions and aggregate capital dynamics generates a fixed-point problem absent when investment is instantaneous.

In short, the agent commits $i_t$ at prices $(r_t, w_t)$ and state $(a_t, k_t, z_t)$, but the payoff-relevant environment at $t+\tau$ differs along all four dimensions. Moreover, the agent's own interim consumption decisions over $[t, t+\tau]$ interact with the locked-up investment to determine her liquid position at the exercise date. This combination of past commitment, interim choices, and interim shocks—inescapable whenever $\tau > 0$—is what reintroduces the Kuhn-Tucker structure into the continuous-time formulation.

## 4.2 The Hybrid Formulation

The full model with time-to-build delay $\tau > 0$ generates a delay differential equation (DDE) for the capital stock. The associated HJB equation lives in an infinite-dimensional state space that includes the entire history of uncommitted investment

over $[t-\tau, t]$ (Fabbri et al., 2017). DDEs are well understood in physics and engineering, where the delayed variable evolves without inequality constraints. In economics, the agent faces a borrowing constraint throughout the gestation interval, so the DDE is coupled with Kuhn-Tucker conditions at every commitment point—a combination that pushes the HJB into infinite-dimensional state space with inequality constraints. The value function is defined on a function space, and existence and uniqueness results require the theory of viscosity solutions in infinite dimensions, which remains an active area of research.

I adopt a hybrid formulation that makes an additional modeling choice: investment commitments occur at discrete dates spaced $\tau$ apart, rather than continuously. This restriction reduces the state space from infinite to finite dimensions while preserving the economic mechanism—the temporal wedge between outlay and payoff—that generates the KT re-emergence. The key simplification is to separate the two controls by their temporal nature:

- **Consumption** $c_t$ is a continuous flow control, chosen at every instant.
- **Investment** $i$ is chosen at discrete commitment points spaced $\tau$ apart.

The agent chooses consumption continuously (as in the standard CT model) but can only commit new investment at dates $0, \tau, 2\tau, 3\tau, \ldots$.[3] Between commitment points, the agent controls only consumption, and the committed investment flows through the delay pipeline toward eventual installation.

The hybrid formulation is a substantive modeling choice that restricts investment opportunities to discrete dates—economically natural, since construction projects are initiated at discrete moments, not continuously. The KT re-emergence result (Proposition 4.4 below) is robust to this choice. The full DDE model also features KT conditions at any commitment point, because the source of the multiplier is the temporal wedge $\tau > 0$, not the discreteness of investment dates.

The model has a natural two-phase structure.

**Phase 1: Flow phase ($s \in (0, \tau)$).** Between commitment points, the agent's only control is consumption. To track position within the commitment cycle, augment the state with a timer variable $s \in [0, \tau]$ that counts time elapsed since the last commitment point. The state vector is $(a, k, z; s)$. During the flow phase, liquid

[3] The restriction to equally spaced commitment points is a simplification. In Online Appendix C, I consider the extension in which the agent chooses both the timing and the magnitude of investment commitments, and show that the main results are robust.

assets evolve as

$$\mathrm{d}a = \big[r\,k + w\,z - c\big]\mathrm{d}t + \sigma_a\,\mathrm{d}W^a, \tag{40}$$

where the investment expenditure $i$ is absent because it was already deducted at the commitment point. Capital depreciates:

$$\mathrm{d}k = -\delta\,k\,\mathrm{d}t. \tag{41}$$

The timer advances deterministically: $\mathrm{d}s = \mathrm{d}t$.

The value function during the flow phase satisfies the HJB equation

$$\rho\,V(a,k,z;s) = \max_{c\geq 0}\left\{u(c) + V_a\big[r\,k + w\,z - c\big] - V_k\,\delta k + V_s + \mathcal{A}_z V + \frac{\sigma_a^2}{2}V_{aa}\right\}, \tag{42}$$

where $\mathcal{A}_z$ denotes the infinitesimal generator of the $z$ process, acting on $V$ through its $z$-derivatives. The first-order condition for consumption is standard:

$$u'(c) = V_a(a,k,z;s). \tag{43}$$

No Kuhn-Tucker multiplier appears in the consumption FOC: as in the standard CT formulation, the borrowing constraint on liquid assets is handled by the boundary condition $V_a(a,k,z;s) \to \infty$ as $a \to \underline{a}^+$ (or equivalently, by a state-constraint boundary condition at $a = \underline{a}$).

The economic content of the flow phase is crucial: during $[t, t+\tau]$, the agent's consumption decisions drain liquid wealth, while the committed investment $i$ remains locked in the pipeline and generates no income. Whether the borrowing constraint binds before the payoff arrives depends on the interaction between the past commitment (which reduced liquid wealth at the commitment point), the agent's interim consumption path (which further draws down liquid assets), and the realization of shocks over the gestation interval. All three feed into the value function $V(a,k,z;s)$, and it is their joint effect that determines the shadow cost of commitment at Phase 2.

**Phase 2: Commitment point ($s = \tau$).** At the commitment point, two events occur simultaneously: (i) investment $i_{\text{prev}}$ committed $\tau$ periods ago arrives and is installed as capital, and (ii) the agent chooses new investment $i \geq 0$. The value

function at the commitment point is

$$V(a,k,z;\tau) = \max_{i\geq 0}\ V\big(a-i,\ k+i_{\text{prev}},\ z;\ 0^+\big) \quad \text{s.t.} \quad a-i \geq \underline{a}, \tag{44}$$

where $i_{\text{prev}}$ is the investment committed at the previous commitment point (which matures now), and $V(\cdot,\cdot,\cdot;0^+)$ is the continuation value at the start of the next flow phase. The pipeline investment $i_{\text{prev}}$ is not an additional state variable: it is determined by the agent's optimal choice at the previous commitment point and remains fixed during the flow phase. In stationary equilibrium, $i_{\text{prev}} = i^*(a',k',z')$, where $(a',k',z')$ is the state at the preceding commitment point, so the commitment-point condition (44) defines a fixed-point problem across successive commitment dates.[4] The constraint $a-i \geq \underline{a}$ ensures that the agent's post-investment liquid assets satisfy the borrowing limit.

This formulation makes transparent the source of the commitment friction. The agent at the commitment point faces a *static* optimization problem: choose $i$ to maximize $V(a-i,k+i_{\text{prev}},z;0^+)$ subject to the inequality constraint $a-i \geq \underline{a}$. This is a constrained maximization problem with an inequality constraint, and its solution generically involves a Kuhn-Tucker multiplier. The continuous-time structure of the flow phase does not eliminate this multiplier. The commitment decision is not a flow; it is a lump-sum decision with consequences that extend over $[t, t+\tau]$.

## 4.3 The HJB System

The complete HJB system consists of three components: the flow-phase HJB equation, the commitment-point optimality condition, and the boundary conditions.

**Flow-phase HJB.** For $s \in (0,\tau)$ and $a > \underline{a}$:

$$\rho\, V = \max_{c\geq 0}\left\{u(c) + V_a\big[r\,k + w\,z - c\big] - V_k\,\delta k + V_s + \mathcal{A}_z V + \frac{\sigma_a^2}{2}V_{aa}\right\}. \tag{45}$$

The optimal consumption policy is $c^*(a,k,z;s) = (u')^{-1}(V_a)$. This equation is a standard second-order PDE in the state variables $(a,k,z)$ with an additional first-

[4] The continuation value $V(a-i,k+i_{\text{prev}},z;0^+)$ implicitly depends on the newly committed investment $i$ through the terminal condition of the flow-phase HJB at $s=\tau$, where $i$ matures and augments capital. Thus $V$ is not monotonically decreasing in $i$: the total derivative $dV/di$ reflects both the liquidity cost of reduced post-investment wealth and the future benefit of pipeline capital.

order term in the timer $s$. It contains no Kuhn-Tucker multiplier. The borrowing constraint is handled entirely by the boundary behavior:

$$\lim_{a \to \underline{a}^+} V_a(a, k, z; s) = +\infty \quad \text{for all } (k, z, s), \tag{46}$$

or equivalently, by a state-constraint boundary condition that ensures the optimal consumption policy prevents $a_t$ from crossing $\underline{a}$. This is precisely the mechanism identified by Achdou et al. (2022): the flow control (consumption) can be adjusted at every instant to steer the state away from the boundary, so the constraint is enforced through the value function's curvature rather than through a multiplier.

**Commitment-point condition.** At $s = \tau$, the agent solves (44). The Lagrangian for this problem is

$$\mathcal{L}(i, \mu) = V\big(a - i,\, k + i_{\text{prev}},\, z;\, 0^+\big) + \mu\big(a - i - \underline{a}\big), \tag{47}$$

with first-order and complementary slackness conditions:

$$V_a\big(a - i^*,\, k + i_{\text{prev}},\, z;\, 0^+\big) = \mu, \tag{48}$$

$$\mu \geq 0, \tag{49}$$

$$\mu\big(a - i^* - \underline{a}\big) = 0. \tag{50}$$

If the constraint is not binding ($a - i^* > \underline{a}$), then $\mu = 0$ and the optimal investment satisfies $V_a(a - i^*, k + i_{\text{prev}}, z; 0^+) = 0$ when an interior solution exists, or is determined by the corner $i^* = 0$ when the agent prefers no investment. More precisely, define the unconstrained optimum $i^{\text{unc}}$ as the solution to $V_a(a - i, k + i_{\text{prev}}, z; 0^+) = 0$ when such a solution exists in $[0, a - \underline{a}]$. The constrained optimum is

$$i^* = \min\big\{i^{\text{unc}},\ a - \underline{a}\big\}^+, \tag{51}$$

where $x^+ \equiv \max\{x, 0\}$. When $i^* = a - \underline{a}$, the multiplier is $\mu = V_a(\underline{a}, k + i_{\text{prev}}, z; 0^+) > 0$, and the complementary slackness condition (50) is satisfied with equality.

**Boundary conditions.** The system is closed by:

(i) The state-constraint boundary at $a = \underline{a}$ during the flow phase: $V$ satisfies the

HJB with the drift of $a$ nonnegative at the boundary (the agent never borrows beyond $\underline{a}$).

(ii) Matching conditions at $s = 0$ and $s = \tau$: the value function is continuous across commitment points, $V(a, k, z; \tau^-) = V(a, k, z; \tau)$, and the commitment-point condition (44) connects $V(\cdot; \tau)$ to $V(\cdot; 0^+)$.

(iii) Boundary conditions on $z$: determined by the specific idiosyncratic productivity process (absorbing barriers, reflecting barriers, or natural boundaries, depending on the process).

(iv) Transversality: $\lim_{t\to\infty} e^{-\rho t}\mathbb{E}[V(a_t, k_t, z_t; s_t)] = 0$.

*Remark* 4.2. The two-phase structure reveals an asymmetry in how the borrowing constraint is enforced across the two controls. During the flow phase, consumption is adjusted continuously, and the constraint is handled by the boundary behavior of $V$—no multiplier is needed, consistent with the AHLLM insight. At the commitment point, investment is a lump-sum decision, and the constraint is enforced by the Kuhn-Tucker multiplier $\mu$—exactly as in discrete-time models. The continuous-time formulation with $\tau > 0$ thus requires *both* enforcement mechanisms simultaneously.

## 4.4 Re-emergence of Kuhn-Tucker

The central result of this section is that the Kuhn-Tucker multiplier $\mu$ in (48)–(50) is *not* absorbed into the boundary condition on $V$. This stands in direct contrast to the standard CT formulation, where the absence of commitment frictions allows all inequality constraints to be handled by the value function's boundary behavior.

To see why absorption fails, consider an agent at a commitment point with liquid assets $a > \underline{a}$. In the standard CT model (with $\tau = 0$), such an agent is strictly interior to the feasible region and faces no binding constraint: the borrowing limit is handled by the continuous adjustment of consumption as $a$ approaches $\underline{a}$. With $\tau > 0$, the same agent—despite having $a > \underline{a}$—may face a binding constraint at the commitment point. The reason is that committing $i$ reduces post-investment assets to $a - i$. The agent must then sustain the borrowing constraint $a_s \geq \underline{a}$ for all $s \in [t, t + \tau]$ starting from $a - i$ rather than $a$. Whenever $\sigma_a > 0$ or the income process $w_s z_s$ is uncertain, there is a positive probability that $a_s$ hits $\underline{a}$ during $[t, t + \tau]$. In this case the agent may optimally choose $i^* < a - \underline{a}$, and the multiplier $\mu$ will be positive even though the constraint at the commitment point is not binding.

Table 2: Constraint handling: standard CT vs. commitment model

| | **Standard CT** (Achdou et al., 2022) | **CT with commitment** ($\tau > 0$) |
|---|---|---|
| Consumption FOC | $u'(c) = V_a$ (no KT) | $u'(c) = V_a$ (no KT) |
| Investment FOC | $f'(i) = 1$ (Tobin's $q = 1$) | $V_a = \mu \geq 0$, with CS |
| Constraint handling | Boundary on $V$ | Boundary (Phase 1) + KT (Phase 2) |

*Notes*: CS denotes complementary slackness. KT denotes Kuhn-Tucker conditions.

More precisely, $\mu$ captures the shadow cost of commitment: the expected cost of resources locked up during gestation, inaccessible for meeting liquidity needs. This shadow cost is positive whenever the borrowing constraint has a positive probability of binding during $[t, t + \tau]$—which occurs for a set of states with positive measure whenever $\tau > 0$ and the environment is stochastic.

The following table summarizes the contrast between the standard CT formulation and the commitment model.

In the standard CT model, the investment FOC is determined by the production technology: marginal product of investment equals one, and Tobin's $q$ is identically one in the absence of adjustment costs (Hayashi, 1982). The agent's wealth and the borrowing constraint do not enter the investment decision. With commitment frictions, the investment FOC acquires a Kuhn-Tucker multiplier that depends on the agent's wealth, the proximity to the borrowing constraint, and the length of the commitment period $\tau$. Investment becomes wealth-determined, echoing the result from the three-date example in Section 2.

*Remark* 4.3. The multiplier $\mu$ prices the *commitment cost*—the loss of liquidity during the gestation period—and should not be confused with a risk premium. A risk premium depends on the covariance between the investment return and the agent's marginal utility; the commitment cost depends on the temporal gap between outlay and payoff. The two interact in stochastic environments but are conceptually distinct.

## 4.5 The Deterministic Case

For clarity, consider the simplest version of the model: a deterministic economy with $\sigma_a = \sigma_z = \sigma_Z = 0$, where prices evolve along a known transition path. In the standard

CT model ($\tau = 0$), investment is pinned down by technology: $f'(i) = 1$, and the KT multiplier is zero whenever $a > \underline{a}$.

With $\tau > 0$, the KT multiplier can be strictly positive, provided the economy is on a non-trivial transition path. The mechanism is as follows. Suppose the economy is converging to steady state, with interest rate $r_t$ rising over time ($\dot{r}_t > 0$). An agent who commits investment $i_t$ at date $t$ locks in current resources but receives capital income at the higher rate $r_{t+\tau}$. This is advantageous in terms of investment returns, but it also means that during $[t, t+\tau]$, the agent's liquid assets are reduced by $i_t$ while the offsetting capital income has not yet materialized. If the agent's income flow during gestation is insufficient to cover consumption—because $w_t z_t$ is low or interest payments $r_t k_t$ are modest—she may hit the borrowing constraint before the investment matures.

The commitment cost arises from the *timing mismatch* between outlay and payoff: the agent cannot access the future income stream during the gestation period, and the committed resources are illiquid for the duration $\tau$.

Formally, in the deterministic case the commitment-point condition becomes

$$V(a, k, z; \tau) = \max_{i \geq 0} \; V\big(a - i,\, k + i_{\text{prev}},\, z;\, 0^+\big) \quad \text{s.t.} \quad a_s \geq \underline{a} \;\; \forall\, s \in [t, t+\tau], \tag{52}$$

where the constraint now requires that the *entire path* of liquid assets over the gestation period satisfies the borrowing limit, not just the initial post-investment level. Under perfect foresight, the agent can compute this path exactly. The Kuhn-Tucker multiplier $\mu$ is positive whenever the path would violate the constraint for the unconstrained investment level $i^{\text{unc}}$, even if $a - i^{\text{unc}} > \underline{a}$ at the commitment point itself.

The commitment cost operates through the Kuhn-Tucker multiplier and has no analogue in the standard CT formulation. It is conceptually distinct from the option value of waiting (Abel and Eberly, 1996). The option value arises from uncertainty about future returns; the commitment cost arises from the temporal mismatch between outlay and payoff. In stochastic environments, the two interact—uncertainty amplifies the commitment cost by increasing the probability that the borrowing constraint binds during gestation.

## 4.6 Formal Propositions

The preceding analysis motivates four formal results. KT re-emergence holds generically (Proposition 4.4), including in deterministic environments (Proposition 4.5). The delay parameter $\tau$ connects CT and DT as endpoints of a single family (Proposition 4.6). The commitment channel has a definite sign for aggregate investment (Proposition 4.7). Full proofs are in Online Appendix A.

**Proposition 4.4** (Re-emergence of Kuhn-Tucker)**.** *In the continuous-time model with time-to-build delay $\tau > 0$, the optimal investment at each commitment point satisfies the complementary slackness conditions (48)–(50) with Kuhn-Tucker multiplier $\mu(a, k, z) \geq 0$. The multiplier is strictly positive on a set $\mathcal{S} \subset \{(a, k, z) : a > \underline{a}\}$ with positive measure whenever $\tau > 0$ and either $\sigma_a > 0$ or $\sigma_Z > 0$.*

*Proof.* At the commitment point, the agent solves a static optimization problem with an inequality constraint. Standard results from convex optimization yield the KT conditions (48)–(50). To show that $\mu > 0$ on a set of positive measure, consider an agent with $a$ slightly above $\underline{a}$ and $\sigma_a > 0$. The unconstrained optimal investment $i^{\text{unc}} > 0$ would reduce post-investment assets to $a - i^{\text{unc}} < \underline{a}$, violating the constraint. The constrained optimum $i^* = a - \underline{a}$ entails $\mu = V_a(\underline{a}, k + i_{\text{prev}}, z; 0^+) > 0$. Continuity of the value function ensures that $\mu > 0$ extends to a neighborhood of positive measure. The full proof, which establishes the result for all states in $\mathcal{S}$, proceeds by showing that the probability of the borrowing constraint binding during $[t, t + \tau]$ is positive for a set of initial conditions with positive measure. See Online Appendix A. ■

**Proposition 4.5** (Deterministic Kuhn-Tucker)**.** *In the deterministic economy ($\sigma_a = \sigma_z = \sigma_Z = 0$), if the economy is on a transition path with $\dot{r}_t \neq 0$ or $\dot{w}_t \neq 0$ over $[t, t + \tau]$, there exist states $(a, k, z)$ with $a > \underline{a}$ at which the Kuhn-Tucker multiplier $\mu > 0$.*

*Proof.* Under perfect foresight, the agent at the commitment point can compute the entire path of liquid assets over $[t, t + \tau]$. When prices are changing ($\dot{r}_t \neq 0$ or $\dot{w}_t \neq 0$), the income flow during the gestation period differs from the income flow at the commitment point. For agents with liquid assets $a$ close to $\underline{a}$, the unconstrained investment $i^{\text{unc}}$ would cause the asset path to violate $a_s \geq \underline{a}$ at some $s \in (t, t + \tau]$, even though $a - i^{\text{unc}} \geq \underline{a}$ at the commitment point. The binding constraint during the gestation period makes $\mu > 0$ at the commitment point. See Online Appendix A. ■

**Proposition 4.6** (Convergence and Interpolation)**.** *The commitment model with delay parameter $\tau$ indexes a continuous family of models with the following properties:*

*(i) As $\tau \to 0$, the value function, optimal policies, and stationary distribution of the commitment model converge to those of the standard continuous-time model (Achdou et al., 2022).*

*(ii) As $\tau \to \Delta t$ for a fixed period length $\Delta t > 0$, the commitment model converges to a discrete-time model with period length $\Delta t$.*

*(iii) For $\tau \in (0, \Delta t)$, the model interpolates continuously between the CT and DT formulations.*

*The delay parameter $\tau$ thus indexes a continuous family interpolating between the two formulations.*

*Proof.* For (i): as $\tau \to 0$, the commitment points become dense, the flow phase shrinks to zero, and the commitment-point condition (44) converges to the standard continuous-time FOC. The KT multiplier $\mu \to 0$ uniformly because the probability of the constraint binding during an interval of length $\tau \to 0$ vanishes. For (ii): as $\tau \to \Delta t$, the flow phase spans the full period, and the commitment-point condition coincides with the Euler equation of the discrete-time model. For (iii): continuity follows from the continuous dependence of the value function on the parameter $\tau$ via the maximum theorem and the continuous dependence of the diffusion on its drift and volatility parameters. See Online Appendix A. ■

**Proposition 4.7** (Precautionary Under-Investment)**.** *Compared to the standard continuous-time model ($\tau = 0$), the commitment model with $\tau > 0$ features:*

*(i) Lower average investment: $\mathbb{E}[i^*(\tau)] < \mathbb{E}[i^*(0)]$.*

*(ii) Higher average liquid asset holdings: $\mathbb{E}[a^*(\tau)] > \mathbb{E}[a^*(0)]$.*

*(iii) A steeper investment policy near the borrowing constraint: $|\partial i^*/\partial a|$ is larger near $a = \underline{a}$ under $\tau > 0$ than under $\tau = 0$.*

*All three effects are monotonically increasing in $\tau$.*

*Proof.* The commitment friction reduces optimal investment through the Kuhn-Tucker multiplier: for any state in $\mathcal{S}$, the agent invests strictly less than the unconstrained optimum. In aggregate, this reduces average investment (i). The reduction in investment redirects resources toward liquid asset accumulation, raising average liquid holdings (ii). Near the borrowing constraint, the shadow cost of commitment is highest (the probability of the constraint binding during $[t, t + \tau]$ is largest), so the

investment policy falls off sharply as $a \to \underline{a}^+$, steepening the policy function (iii). Monotonicity in $\tau$ follows because a longer commitment period increases the exposure along all four channels, raising the probability of constraint binding and hence the KT multiplier. See Online Appendix A. ■

*Remark* 4.8. Proposition 4.6 makes precise the equivalence at the heart of the paper. Standard discrete time with period $\Delta t$ implicitly embeds a time-to-build lag of $\Delta t$: investment paid at $t$ yields capital at $t + \Delta t$, and the KT multiplier prices the agent's exposure during that interval. The CT limit $\Delta t \to 0$ sends this lag to zero, eliminating the KT conditions. This elimination is correct if the lag is a discretization artifact—but not if it reflects a structural friction with duration $\tau > 0$. Setting $\tau$ explicitly restores exactly the KT structure that the CT limit removed, while preserving continuous consumption adjustment and all other computational advantages of the CT framework. The parameter $\tau$ is not a grid choice; it is a physical quantity—months of construction, quarters of contract duration—that determines whether the model includes or excludes the commitment cost.

*Remark* 4.9. The precautionary under-investment in Proposition 4.7 is conceptually distinct from the precautionary saving motive in the standard incomplete-markets literature. In the standard model, precautionary saving arises from the convexity of the marginal utility function and operates through the consumption-savings margin. In the commitment model, precautionary under-investment arises from the commitment cost and operates through the investment margin. The two channels are complementary. The standard precautionary saving motive is present in the commitment model and operates during the flow phase. The commitment cost channel operates at commitment points and provides an additional source of under-investment that is absent when $\tau = 0$.

# 5 The HJB-KT Form

The analysis of Section 4 develops the commitment friction for a specific application: time-to-build investment with a borrowing constraint. But the underlying structure is general. Any continuous-time model in which some decisions are instantaneous and others involve commitment—locking in resources against a delayed payoff—admits the same two-phase optimality structure. This section provides the complete formulation.

Consider an agent with state $x \in \mathbb{R}^n$, subject to a constraint $x \in \mathcal{C} \subseteq \mathbb{R}^n$ (a borrowing limit, a short-sale constraint, a regulatory bound). The agent makes two types of decisions. *Flow controls* $c$—consumption, portfolio rebalancing, pricing—are adjusted continuously and take effect instantaneously. *Commitment controls* $\iota \geq 0$—investment, durable purchases, contract initiation—are chosen at discrete dates $0, \tau, 2\tau, \ldots$ and locked in for a gestation period of duration $\tau > 0$. Between commitment dates, the agent's only instrument is the flow control. The commitment control, once chosen, flows through the pipeline toward eventual realization: the expenditure is sunk, the resources are illiquid, and the payoff arrives only at the end of the gestation interval. Let $s \in [0, \tau]$ denote the time elapsed since the last commitment point.

**Definition 5.1** (The HJB-KT System)**.** The value function $V(x; s)$ satisfies the following coupled system.

**(I) Flow-phase HJB** *(between commitment dates: $s \in (0, \tau)$, $x \in \mathrm{int}(\mathcal{C})$).*

Between commitment dates, the agent optimizes over the flow control alone:

$$\rho\, V(x; s) \;=\; \max_{c} \Big\{ u(c, \bar{\iota}) \;+\; \mathcal{L}^{c,\bar{\iota}}\, V(x; s) \;+\; V_s(x; s) \Big\}, \tag{53}$$

where $\bar{\iota}$ is the commitment control chosen at the preceding commitment point (a fixed parameter throughout the flow phase), $\mathcal{L}^{c,\bar{\iota}}$ is the infinitesimal generator of the state process under controls $(c, \bar{\iota})$, and $V_s$ captures the deterministic advance of the timer toward the next commitment date. The constraint $x \in \mathcal{C}$ is enforced entirely through the boundary behavior of $V$: the marginal value $\|\nabla V\| \to \infty$ as $x \to \partial\mathcal{C}$, so the optimal flow control steers the state away from the boundary before it is reached. Because the flow control adjusts instantaneously, it can always prevent the state from crossing the constraint. No Kuhn–Tucker multiplier appears.

**(II) Commitment-point condition** *(at commitment dates: $s = \tau$).*

At a commitment date, any prior commitment matures (its payoff is realized) and the agent initiates a new commitment $\iota \geq 0$. The agent solves a constrained static optimization:

$$V(x; \tau) \;=\; \max_{\iota \geq 0}\; V\big(\varphi(x, \iota);\, 0^+\big) \qquad \text{subject to} \quad g(x, \iota) \leq 0, \tag{54}$$

where $\varphi(x, \iota)$ is the post-commitment state transition—encoding the immediate re-

source cost of $\iota$ and the maturation of any prior pipeline commitment—and $g(x, \iota) \leq 0$ requires that the agent can sustain the constraint $x_t \in \mathcal{C}$ throughout the upcoming gestation interval $[t, t + \tau]$.

This is the decisive departure from the standard formulation. The commitment control cannot be revised once chosen, and its resource cost is borne immediately while its payoff arrives only after $\tau$ periods. No infinitesimal adjustment of the flow control can undo a commitment that pushes the state toward the constraint boundary during gestation. The constraint must therefore be enforced by a Kuhn–Tucker multiplier. The Lagrangian and KKT conditions are:

$$\mathcal{L}(\iota, \mu) \;=\; V\big(\varphi(x, \iota);\, 0^+\big) \;-\; \mu \cdot g(x, \iota), \tag{55}$$

$$\nabla_\iota V\big(\varphi(x, \iota^*);\, 0^+\big) \;=\; \mu\, \nabla_\iota g(x, \iota^*), \tag{56}$$

$$\mu \;\geq\; 0, \tag{57}$$

$$\mu \cdot g(x, \iota^*) \;=\; 0. \tag{58}$$

The multiplier $\mu(x) \geq 0$ prices the *commitment cost*: the shadow value of the liquidity locked in the pipeline and inaccessible for consumption smoothing or self-insurance during gestation.

**(III) Matching conditions.**

The value function is continuous across the commitment boundary:

$$V(x;\, \tau^-) \;=\; V(x;\, \tau), \tag{59}$$

and the commitment-point condition (54) maps $V(\cdot\,; \tau)$ to $V(\cdot\,; 0^+)$, closing the recursive structure: the flow-phase HJB propagates the value function from $s = 0^+$ forward to $s = \tau$, and the commitment-point condition resets the timer and connects $V(\cdot\,; \tau)$ back to $V(\cdot\,; 0^+)$.

**(IV) Boundary and terminal conditions.**

(a) *State-constraint boundary:* At $x \in \partial\mathcal{C}$ during the flow phase, the drift of $x$ under the optimal flow control is non-negative toward the interior (reflecting boundary), preventing the state from exiting $\mathcal{C}$.

(b) *Transversality:* $\lim_{t\to\infty} e^{-\rho t}\, \mathbb{E}\big[V(x_t;\, s_t)\big] = 0$.

I call this the *HJB-KT form*: the value function is characterized by a Hamilton–

Jacobi–Bellman equation between commitment dates and Kuhn–Tucker conditions at commitment dates.

*Remark* 5.2. The delay parameter $\tau$ indexes a continuous family of models:

(i) *Standard continuous time* ($\tau \to 0$). The flow phase shrinks to a point, commitment dates become dense in $\mathbb{R}_+$, and the commitment control becomes a second flow control. The probability of violating the constraint during an interval of vanishing length tends to zero, so the KT multiplier $\mu \to 0$ uniformly. The system collapses to a single HJB equation with boundary conditions—the formulation of Achdou et al. (2022). Phase II is absent entirely.

(ii) *Standard discrete time* ($\tau = \Delta t$). The flow phase spans one full period. Phase I collapses to a single step, and the commitment-point condition (54) reduces to the one-period Bellman equation. The KKT conditions (56)–(58) reproduce the discrete-time Euler equation with its Kuhn–Tucker multiplier. Phase I is absent.

(iii) *Interpolation* ($\tau \in (0, \Delta t)$). Both phases are active. The KT multiplier increases monotonically from 0 to the discrete-time multiplier as $\tau$ increases from 0 to $\Delta t$.

The CT–DT choice is therefore not merely a matter of numerical convenience but a substantive modeling decision: it determines the value of $\tau$ and hence the size of the commitment cost that the model includes or excludes.

*Remark* 5.3. A single constraint—$x \in \mathcal{C}$, imposed at all dates—is enforced by two fundamentally different mechanisms operating simultaneously. During the flow phase, the constraint is handled by the curvature of the value function: $\|\nabla V\| \to \infty$ at the boundary drives optimal consumption toward zero, ensuring non-negative drift. This is the Achdou et al. (2022) insight. At commitment points, the constraint is handled by the Kuhn–Tucker multiplier $\mu$: committing $\iota$ discretely reduces resources, and no instantaneous flow adjustment can undo this reduction before the constraint is evaluated. Standard continuous-time models (all controls instantaneous) use only the first mechanism. Standard discrete-time models (all controls periodic) use only the second. The HJB-KT form requires both.

A full characterization of the HJB-KT system—existence and uniqueness of solutions, regularity properties, and convergence of numerical schemes—is beyond the scope of this paper. The mathematical structure couples a PDE (the HJB equation in Phase 1) with an optimization problem subject to inequality constraints (Phase 2).

The timer variable $s$ introduces a backward-in-time dimension linking the two phases. I flag this as an important direction for future research. The results of Fabbri et al. (2017) on HJB equations in infinite-dimensional spaces and the viscosity solution framework of Barles and Souganidis (1991) provide natural starting points for such an analysis.

# 6 Quantitative Analysis

This section asks: how large are the effects? I embed the commitment friction in a calibrated general equilibrium heterogeneous-agent economy. For empirically relevant gestation lags, the commitment channel lowers the aggregate capital stock by 2–8%, raises the equilibrium interest rate by 20–79 basis points, and transforms the investment policy function from technology-determined to wealth-determined.

## 6.1 Model

The economy is populated by a continuum of agents of measure one. Each agent is endowed with one unit of labor, supplied inelastically, and faces idiosyncratic productivity $z_t$ that follows an Ornstein-Uhlenbeck process,

$$\mathrm{d}z_t = -\theta_z(z_t - \bar{z})\,\mathrm{d}t + \sigma_z\,\mathrm{d}W_t, \tag{60}$$

where $\bar{z}$ is the unconditional mean, $\theta_z$ governs mean reversion, and $\sigma_z$ is the volatility. An agent with productivity $z$ earns labor income $wz$, where $w$ is the equilibrium wage.

Each agent holds liquid wealth $a_t$ and may own capital $k_t$. Both liquid wealth and installed capital earn the equilibrium return $r = \alpha A K^{\alpha-1} L^{1-\alpha} - \delta$ (the net marginal product of capital), and liquid wealth is subject to the borrowing constraint $a_t \geq \underline{a}$. The agent chooses consumption $c_t$ and, at equally spaced commitment dates, the scale of new investment projects $i$. Investment is subject to a time-to-build lag $\tau \geq 0$: a project initiated at date $t$ costs $i$ immediately and delivers $i$ units of productive capital at date $t+\tau$. During the interval $[t, t+\tau]$, the committed resources are illiquid—they reduce the agent's liquid wealth at inception but do not generate income or serve as collateral until installation is complete.

The individual state is $(a, k, z, s)$, where $s \in [0, \tau]$ is the time elapsed since the last commitment point, as in Section 4. For $s \in (0, \tau)$, the agent is in Phase 1 (gestation):

she chooses only consumption, the HJB equation governs the flow phase, and the borrowing constraint binds through a standard boundary condition. At $s = \tau$, the agent is in Phase 2 (commitment): she chooses a new project scale $i \geq 0$, which must satisfy the Kuhn-Tucker conditions derived in Section 4. The aggregate production function is

$$Y = F(K, L) = AK^{\alpha}L^{1-\alpha}, \tag{61}$$

where $K$ is aggregate capital and $L = \int z \, \mathrm{d}\Phi(z)$ is aggregate effective labor.

The competitive equilibrium consists of prices $(r, w)$, individual policy functions $(c, i)$, and a stationary distribution $\Phi(a, k, z, s)$ such that (i) agents optimize given prices, (ii) the representative firm maximizes profit, and (iii) markets clear:

$$K = \int k \, \mathrm{d}\Phi, \qquad L = \int z \, \mathrm{d}\Phi, \qquad r = \alpha AK^{\alpha-1}L^{1-\alpha} - \delta, \qquad w = (1-\alpha)AK^{\alpha}L^{-\alpha}. \tag{62}$$

The model nests the standard continuous-time benchmark when $\tau = 0$, in which case investment is instantaneous and the Phase 2 Kuhn-Tucker conditions are absent.

## 6.2 Calibration

The model is calibrated at quarterly frequency. Table 3 reports the parameter choices.

Preferences are CRRA, $u(c) = c^{1-\gamma}/(1-\gamma)$, with coefficient $\gamma = 2$ and discount rate $\rho = 0.01$, implying an annual discount factor of approximately 0.96. The exogenous wealth shock $\sigma_a$ from the general formulation (36) is set to zero; all idiosyncratic risk operates through the productivity channel. The production function has capital share $\alpha = 0.36$ and depreciation rate $\delta = 0.025$ per quarter (approximately 10% per annum). TFP is normalized to $A = 1$. The income process parameters $\theta_z = 0.50$ and $\sigma_z = 0.20$ imply a quarterly autocorrelation of $e^{-\theta_z} \approx 0.61$ for the productivity level, broadly consistent with the estimates used by Aiyagari (1994).[5] The borrowing limit is set to $\underline{a} = 0$ (natural borrowing limit).

The key parameter is the time-to-build lag $\tau$. I consider four values: $\tau = 0$ (no lag, the standard CT benchmark), $\tau = 1$ quarter, $\tau = 2$ quarters, and $\tau = 4$ quarters. The empirical literature on investment gestation lags provides guidance for this range.

[5] Aiyagari's discrete-time specification uses a quarterly autocorrelation of approximately 0.96 for log income. The OU process in (60) is specified in levels with higher mean reversion, implying faster mixing. The results are not sensitive to this choice within the empirically relevant range.

Table 3: Calibration

| Parameter | Description | Value | Target / Source |
|---|---|---|---|
| *Preferences* | | | |
| $\rho$ | Discount rate | 0.01 | Annual $\beta \approx 0.96$ |
| $\gamma$ | Risk aversion (CRRA) | 2.0 | Standard |
| $\alpha$ | Capital share | 0.36 | Labor share = 0.64 |
| $\delta$ | Depreciation rate | 0.025 | Quarterly, 10% annual |
| $A$ | TFP | 1.0 | Normalization |
| $\sigma_z$ | Volatility of $z$ | 0.20 | Aiyagari (1994) |
| $\theta_z$ | Mean-reversion of $z$ | 0.50 | Quarterly $\rho_z = e^{-\theta_z} \approx 0.61$ |
| $\underline{a}$ | Borrowing limit | 0.0 | Natural borrowing limit |
| $\tau$ | Time-to-build lag | $\{0, 1, 2, 4\}$ | Quarters; see text |

*Notes:* The table reports the baseline quarterly calibration. The time-to-build parameter $\tau$ is varied across experiments. $\tau = 0$ corresponds to the standard continuous-time benchmark. Empirical estimates of investment gestation lags range from 1 to 8 quarters depending on the sector (Koeva, 2000; Montgomery, 1995).

Koeva (2000) documents that the median time to build for nonresidential structures in the United States is approximately 12 months, with substantial variation across sectors: light manufacturing facilities require 4–6 months, heavy industrial plants 12–18 months, and power generation facilities 24–36 months. Montgomery (1995) reports similar estimates using Census Bureau data on construction completion patterns, finding median lags of 1–2 quarters for equipment, 3–4 quarters for commercial buildings, and 6–8 quarters for large infrastructure. The range $\tau \in \{1, 2, 4\}$ quarters spans the empirically relevant region for private-sector capital investment.

In addition to the three CT models with $\tau \in \{0, 1, 2, 4\}$, I solve a standard discrete-time (DT) model at quarterly frequency as a reference point. The DT model is the Aiyagari (1994) economy with the same parameters, solved using standard value function iteration with Euler equation methods. Comparing the DT model to the CT models allows assessment of whether commitment frictions in CT produce dynamics that resemble the DT economy, as the theoretical results suggest.

## 6.3 Computation

I briefly describe the computational approach; full details are in Online Appendix B. The state space is discretized on a grid over $(a, k, z)$ with the timer variable $s$ discretized on a uniform grid over $[0, \tau]$. For Phase 1 ($s \in (0, \tau)$), the HJB equation is solved using an upwind finite difference scheme, stepping forward in $s$ from the initial condition at $s = 0^+$ (the start of the flow phase). At each step, the consumption policy is obtained from the first-order condition $u'(c) = V_a$, and the borrowing constraint is enforced through a state-constraint boundary condition at $a = \underline{a}$.

At commitment points ($s = \tau$), the agent solves a constrained optimization problem over the investment scale $i$, subject to the requirement that liquid wealth remains above $\underline{a}$ throughout the upcoming gestation interval. This problem involves the Kuhn-Tucker conditions derived in Proposition 4.4: the multiplier on the commitment-period borrowing constraint prices the probability of hitting the borrowing limit during gestation, and it enters the investment first-order condition as a wedge between the marginal cost and the discounted marginal product of capital.

The stationary distribution $\Phi(a, k, z, s)$ is computed from the Kolmogorov forward (KF) equation associated with the individual dynamics. The KF equation is discretized using the adjoint of the finite difference operator from the HJB equation, ensuring mass conservation. General equilibrium is obtained through an outer loop over the interest rate $r$: given $r$, the wage is determined by firm optimization, the individual problem is solved, the stationary distribution is computed, and the capital market clearing condition $K = \int k \, \mathrm{d}\Phi$ is checked. The outer loop adjusts $r$ using bisection until clearing is achieved to a tolerance of $10^{-6}$.

## 6.4 Results

Table 4 reports the stationary equilibrium outcomes for the five models: the standard CT benchmark ($\tau = 0$), the three CT commitment models ($\tau \in \{1, 2, 4\}$), and the standard DT model.

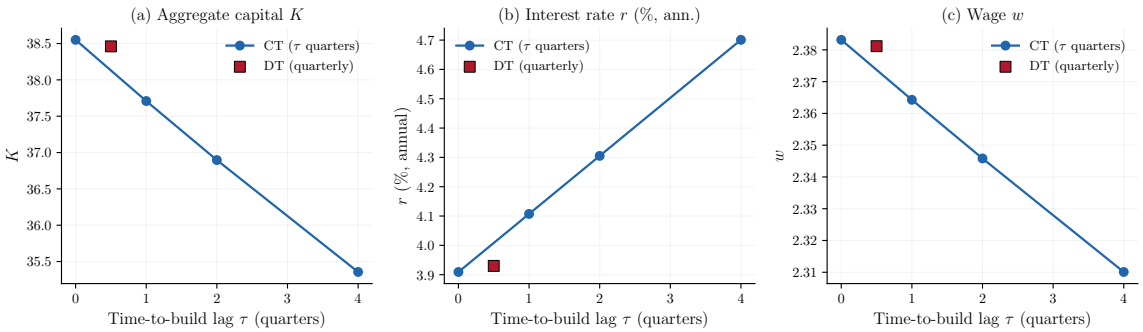
**Aggregate capital and prices.** The commitment friction reduces aggregate capital accumulation. As $\tau$ increases from 0 to 4 quarters, the equilibrium capital stock falls monotonically. The mechanism is the precautionary under-investment channel of Proposition 4.7. Agents who commit resources for a longer gestation period face

Table 4: Stationary equilibrium: Commitment friction and aggregate outcomes

| | Continuous Time | | | | Discrete |
|---|---|---|---|---|---|
| | $\tau = 0$ | $\tau = 1$ | $\tau = 2$ | $\tau = 4$ | Time (DT) |
| Interest rate $r$ (%, ann.) | 3.91 | 4.11 | 4.31 | 4.70 | 3.93 |
| Wage $w$ | 2.38 | 2.36 | 2.35 | 2.31 | 2.38 |
| Capital $K$ | 38.55 | 37.71 | 36.90 | 35.36 | 38.46 |
| Output $Y$ | 3.72 | 3.69 | 3.67 | 3.61 | 3.72 |
| Wealth Gini | 0.34 | 0.33 | 0.33 | 0.33 | 0.34 |
| Top 10% wealth share (%) | 23.9 | 23.8 | 23.8 | 23.3 | 24.3 |

*Notes*: $\tau$ is the time-to-build lag in quarters; DT is the Aiyagari (1994) model at quarterly frequency. In all specifications, $I/K = \delta = 0.025$ in steady state.

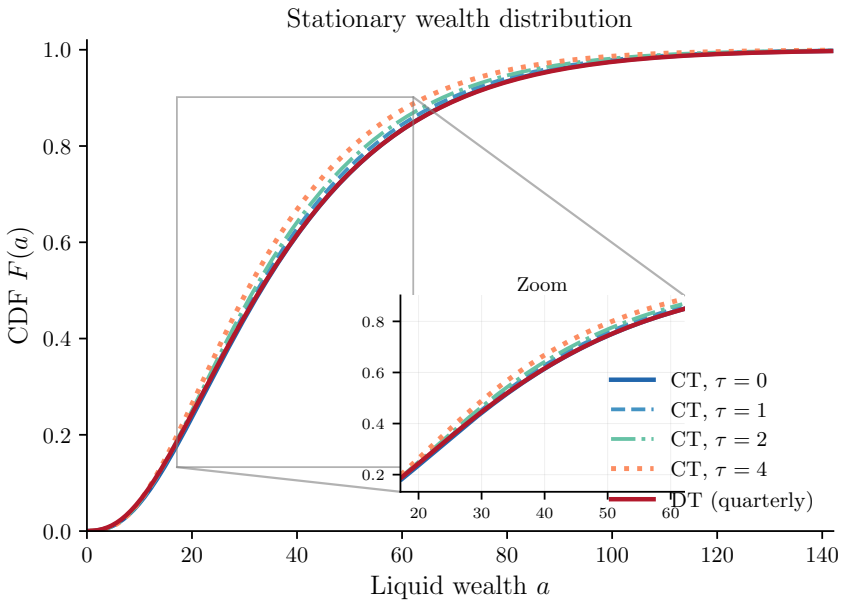
Figure 2: Aggregate outcomes as a function of the time-to-build lag $\tau$




greater exposure to the borrowing constraint during the gap interval and respond by investing less. The reduction in capital raises the equilibrium interest rate and lowers the wage, partially offsetting but not eliminating the decline. Figure 2 displays the paths of $K$, $r$, and $w$ as functions of $\tau$.

**Investment-to-capital ratio.** In stationary equilibrium, $I/K = \delta = 0.025$ regardless of $\tau$, because replacement investment must offset steady-state depreciation in every model. The commitment friction therefore operates entirely on the *level* of capital, not on the replacement rate. Agents invest less in aggregate, accumulate less capital, and replace it at the same proportional rate. The wedge created by the commitment cost shows up in the equilibrium capital stock $K$ and the prices $(r, w)$, not in the steady-state $I/K$ ratio.

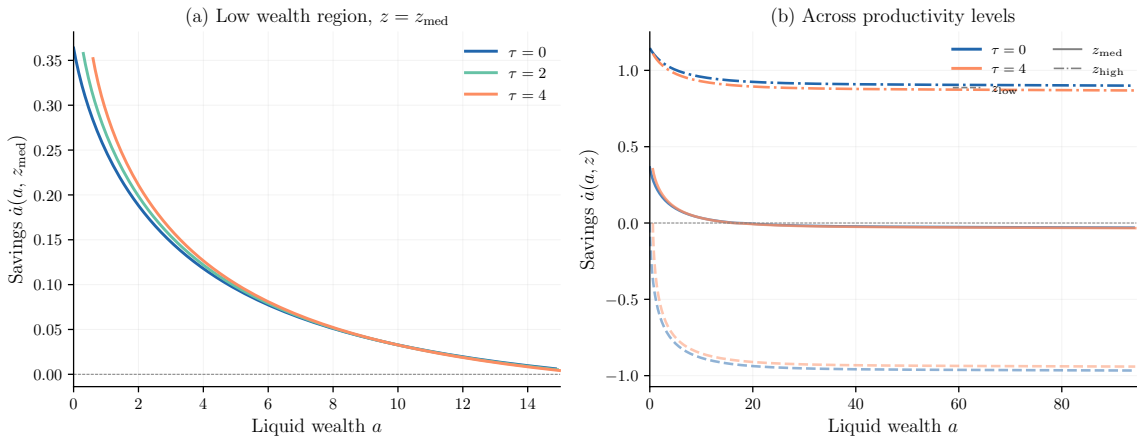
Figure 3: Stationary wealth distribution across models




**Wealth distribution.** The commitment friction has measurable effects on the stationary wealth distribution. Figure 3 plots the cumulative distribution of liquid wealth for the five models. Three features stand out. First, no agents are observed at the borrowing constraint in any specification. The precautionary motive induced by the commitment cost keeps all agents strictly above $\underline{a}$, even though the constraint shapes optimal behavior. Second, the wealth Gini falls slightly with $\tau$ (from 0.34 to 0.33), and the top 10% wealth share declines (from 23.9% to 23.3%). Both patterns reflect the precautionary channel. Agents accumulate larger liquid buffers to self-insure against gestation-period illiquidity, compressing the lower tail and reducing inequality on net. Third, despite this compression, the overall distribution shifts leftward because the capital stock declines. The reduction in inequality coexists with a decline in aggregate wealth.

**Policy functions.** Figure 4 displays the investment policy function $i(a, z)$ at the commitment point ($s = \tau$) for agents with median productivity, evaluated at $\tau \in \{0, 2, 4\}$. In the $\tau = 0$ benchmark, investment is independent of liquid wealth for all $a$ above a low threshold—the technology-determined rule from the analytical results. At $\tau = 2$ and $\tau = 4$, the investment policy becomes an increasing function

Figure 4: Investment policy functions across commitment lag and productivity




of wealth over a wide range. The slope is steeper for larger $\tau$, reflecting the greater bite of the commitment cost. The qualitative pattern matches the closed-form result from the three-date example (Proposition 2.1). The commitment friction transforms investment from a technology condition to a wealth-dependent liquidity rule. In the quantitative model, this transformation occurs smoothly as $\tau$ increases, with the wealth-dependent region expanding and the technology-determined region shrinking.

**Comparison with discrete time.** The standard DT model at quarterly frequency produces outcomes close to, but not identical to, the CT commitment model with $\tau = 1$ quarter. Specifically, the DT model yields $K = 38.46$, while the CT model with $\tau = 1$ gives $K = 37.71$. The residual gap reflects a difference in the consumption margin. In the DT model, consumption is chosen once per period and held fixed; in the CT commitment model, consumption adjusts continuously within the gestation interval. Continuous adjustment provides better self-insurance during gestation, allowing slightly lower liquid buffers—but this effect is small relative to the commitment friction itself. Both models share the essential feature that the Euler equation constrains the agent over a discrete interval. The KT multiplier in the DT Euler equation is the discrete-time counterpart of the commitment-period multiplier in the CT model with $\tau = \Delta t$. The comparison confirms that the CT–DT choice has economic content. The two formulations disagree on the equilibrium capital stock, interest rate, and wealth distribution, and the disagreement traces to the commitment friction that the DT model implicitly contains.

# 7 Discussion

## 7.1 Physics versus Economics

Continuous-time methods in economics are imported from mathematical physics, where the Markov property holds deeply. A particle undergoing Brownian motion does not commit resources, sign contracts, or face borrowing constraints while waiting for a return. Its state evolves without any temporal wedge between "decision" and "outcome." For economic agents, the situation is fundamentally different. A firm that breaks ground on a factory has committed resources that will be illiquid for the duration of construction; a household that signs a mortgage has locked in payments that cannot be costlessly adjusted. In each case, the decision at date $t$ creates obligations extending over $[t, t + \tau]$, and the agent's liquid state depends on the pipeline of earlier commitments. The Markov property is preserved only if the state is augmented to include this pipeline—which is what the timer variable $s$ in Section 4 accomplishes.

The point is not that continuous-time methods are inapplicable. They remain the right tool for decisions that are genuinely instantaneous: consumption, portfolio rebalancing in liquid markets, pricing without menu costs. The point is that the infinitesimal limit works for physical motion but not for economic commitment. Delay differential equations are standard in physics precisely because the delayed variable evolves without inequality constraints—molecules do not face borrowing limits while waiting for a reaction to complete. Economic agents do. The coupling of delay with Kuhn-Tucker conditions is what makes economic DDEs fundamentally harder than their physical counterparts, and it is why the standard CT approach of absorbing constraints into boundary conditions breaks down when $\tau > 0$.

As noted in the Introduction, this distinction maps onto the dual nature of the "+1" in discrete-time Euler equations: it is either (a) a discretization artifact or (b) a structural delay. The KT multiplier in the DT Euler equation similarly plays two roles. It penalizes overshooting the constraint within one period (type (a), vanishing as $\Delta t \to 0$) and prices the cost of commitment over a structural delay (type (b), persisting). The CT limit eliminates both: appropriately for the first, but also for the second—unless the model explicitly includes the commitment friction.

## 7.2 Implications for the AHLLM Program

The results of this paper are complementary to Achdou et al. (2022), not contradictory. Their sidestepping result is correct for instantaneous controls: consumption, portfolio allocation in liquid markets, and any decision whose implementation is immediate. The contribution here is to identify the boundary of that result. When the control involves commitment—a decision whose consequences extend over a positive interval $\tau > 0$—the Kuhn-Tucker structure re-emerges within the continuous-time formulation itself.

The practical implication is that modelers working in the AHLLM framework should distinguish between instantaneous and commitment controls. For the former, the standard boundary-condition approach applies without modification. For the latter, the HJB equation must be supplemented with Kuhn-Tucker conditions at commitment points, as in Section 4. This distinction matters for any application in which a key decision variable involves positive gestation or implementation lags—firm investment, housing transactions, durable consumption, labor contracts. The extension preserves the computational tractability of the continuous-time approach. Section 7.3 develops these applications in detail.

**Robustness to partial pledgeability.** The baseline model (Assumption 4.1) assumes that pipeline investment is entirely non-pledgeable. In practice, a fraction of investment-in-progress may serve as collateral—for example, through construction loans secured by the partially completed structure. The analysis extends to partial pledgeability: if a fraction $\phi \in [0, 1]$ of pipeline investment relaxes the borrowing constraint, the effective constraint at the commitment point becomes $a - i + \phi\, i \geq \underline{a}$, i.e., $a - (1 - \phi)\, i \geq \underline{a}$. For $\phi = 1$ (full pledgeability), the commitment cost vanishes and the standard CT result obtains. For any $\phi < 1$, the KT multiplier is positive on a set of positive measure, and the qualitative results of Section 4 hold. The parameter $\phi$ scales the magnitude of the commitment cost but does not eliminate it.

## 7.3 Beyond Investment

The time-to-build friction analyzed here belongs to a broader class of commitment frictions sharing the same temporal structure. A decision at date $t$ locks in resources over $[t, t + \tau]$, during which the agent cannot costlessly adjust. The Kuhn-Tucker

re-emergence result (Proposition 4.4) applies to any such friction, regardless of its specific economic interpretation. Several applications merit investigation:

In *durable consumption*, large purchases such as housing and vehicles involve gestation lags between the decision and the point at which the durable delivers services. The commitment cost suppresses durable purchases by constrained agents, potentially contributing to the observed concentration of durable expenditure among wealthier households (Grossman and Laroque, 1990; Berger and Vavra, 2015). In *occupational choice*, switching occupations typically requires retraining, during which the worker earns reduced income. The training lag $\tau$ creates a commitment cost analogous to time-to-build: resources are committed now, returns arrive later, and the worker must self-insure during the gap. *Migration* involves relocation costs, job search, and integration lags; during the transition the migrant's income is reduced and expenditures are elevated, creating a gap period analogous to the investment gestation interval. Finally, *research and development* projects commit resources over a development period before any commercial return materializes; the commitment cost of R&D is especially large because the gestation lag is long (years to decades) and the outcome is uncertain.

In each case, the standard CT formulation with instantaneous controls does not capture the commitment cost channel. Embedding these frictions in the HJB-KT framework provides a unified approach to modeling temporal commitment in continuous time.

# 8 Conclusion

When the standard continuous-time framework is applied to an economy with gestation lags, the liquidity cost of the gap period between commitment and payoff is absent. So is the precautionary channel through which that cost shapes decisions and aggregate outcomes. The deviations are economically meaningful: for a four-quarter build lag—typical of nonresidential construction—the commitment channel lowers the equilibrium capital stock by 8% and raises the interest rate by 79 basis points relative to the frictionless benchmark.

The extension is straightforward. Modelers should distinguish between instantaneous controls—for which the boundary-condition approach of Achdou et al. (2022) applies without modification—and commitment controls, which require Kuhn-Tucker conditions at commitment points. The two-phase formulation of Section 4 provides

a tractable way to do so. More broadly, the $\tau$-interpolation result (Proposition 4.6) shows that the CT–DT choice is not a numerical convenience but a substantive economic decision. It determines whether the model includes or excludes a friction whose duration is measurable from data on construction timelines, contract lengths, and delivery lags.

Several directions remain open. First, a rigorous treatment of the HJB-KT system—existence, uniqueness, and regularity for the coupled PDE and constrained optimization problem—building on Barles and Souganidis (1991) and Fabbri et al. (2017). Second, computational refinements exploiting the two-phase structure: parallelization across the timer variable, adaptive grids near the commitment boundary, and machine learning approximations at commitment points. Third, empirical investigation. The parameter $\tau$ is estimable from microdata on investment gestation lags and construction timelines. The model's predictions for wealth-dependent investment and distributional dynamics provide testable implications. Estimates from Koeva (2000) and Montgomery (1995) suggest substantial cross-sectoral variation in $\tau$, implying that the commitment cost channel matters most in capital-intensive industries with long construction timelines.

Time is ticking—and in macroeconomics, it matters whether the tick is a modeling convenience or an economic reality.

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